

Bilinear Classes: A Structural Framework for Provable Generalization in RL

Talk by: Gaurav Mahajan
(UCSD)

Joint work with:



Simon Du



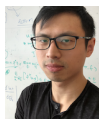
Sham Kakade



Jason Lee



Shachar Lovett



Wen Sun



Ruosong Wang

Progress of RL in practice



- Lots of recent empirical success.

Progress of RL in practice



- Lots of recent empirical success.
- Tackling large state spaces is a central challenge in RL.



- Lots of recent empirical success.
- Tackling large state spaces is a central challenge in RL.
 - Growing theoretical work on assumptions which allow dealing with large state spaces.



- Lots of recent empirical success.
- Tackling large state spaces is a central challenge in RL.
 - Growing theoretical work on assumptions which allow dealing with large state spaces.
 - Can we unify these assumptions?

We aim to understand **natural sufficient conditions** which capture the learnability in a general class of RL models.

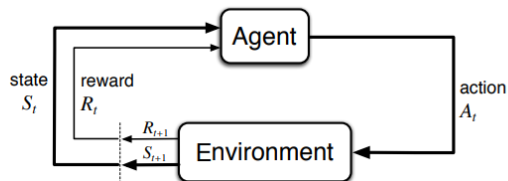
We aim to understand **natural sufficient conditions** which capture the learnability in a general class of RL models.

- **Part I: Generalization in Reinforcement Learning**
Connections to Supervised Learning

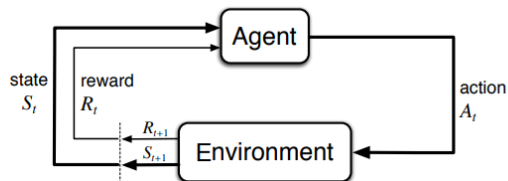
We aim to understand **natural sufficient conditions** which capture the learnability in a general class of RL models.

- **Part I: Generalization in Reinforcement Learning**
Connections to Supervised Learning
- **Part II: Unifying sufficient conditions**
Various model assumptions for generalization in RL
Simple Algorithm and Short Proof

Markov Decision Processes: A Framework for RL

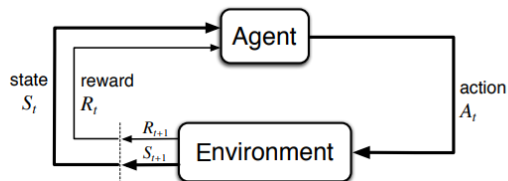


Markov Decision Processes: A Framework for RL



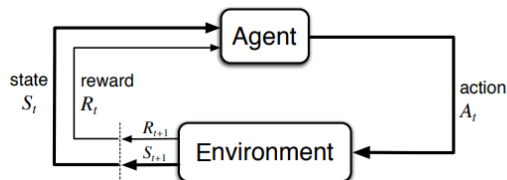
- A **policy** $\pi : \mathcal{S} \rightarrow \mathcal{A}$

Markov Decision Processes: A Framework for RL



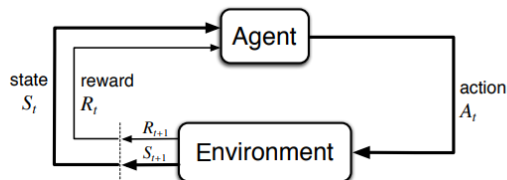
- A **policy** $\pi : \mathcal{S} \rightarrow \mathcal{A}$
 - Mario: Always go right!!

Markov Decision Processes: A Framework for RL



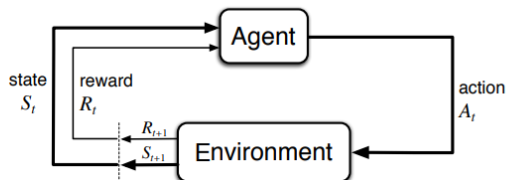
- A **policy** $\pi : \mathcal{S} \rightarrow \mathcal{A}$
 - Mario: Always go right!!
- Execute π to obtain a **H-step trajectory** $s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{H-1}, a_{H-1}, r_{H-1}$

Markov Decision Processes: A Framework for RL



- A **policy** $\pi : \mathcal{S} \rightarrow \mathcal{A}$
 - Mario: Always go right!!
- Execute π to obtain a **H-step trajectory** $s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{H-1}, a_{H-1}, r_{H-1}$
 - Chess: $H \approx 80$, Go: $H = 150$, Dota 2: $H \approx 20000$

Markov Decision Processes: A Framework for RL



- A **policy** $\pi : \mathcal{S} \rightarrow \mathcal{A}$
 - Mario: Always go right!!
- Execute π to obtain a **H-step trajectory** $s_0, a_0, r_0, s_1, a_1, r_1, \dots, s_{H-1}, a_{H-1}, r_{H-1}$
 - Chess: $H \approx 80$, Go: $H = 150$, Dota 2: $H \approx 20000$

Goal

Learn a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ which maximizes $\mathbb{E}_{\pi} \left[\sum_{t=0}^{H-1} r_t \right]$.

Part I: Generalization from Supervised Learning to Reinforcement Learning

Generalization is possible in the IID supervised learning setting!!

Generalization is possible in the IID supervised learning setting!!

To get ϵ -close to best in hypothesis class $\mathcal{F}$, we need # of samples that is:

- Finite Hypothesis class: $O(\log(|\mathcal{F}|)/\epsilon^2)$.

Generalization is possible in the IID supervised learning setting!!

To get ϵ -close to best in hypothesis class $\mathcal{F}$, we need # of samples that is:

- Finite Hypothesis class: $O(\log(|\mathcal{F}|)/\epsilon^2)$.
- Infinite hypothesis classes: $O(\text{VCdim}(\mathcal{F})/\epsilon^2)$.

Generalization is possible in the IID supervised learning setting!!

To get ϵ -close to best in hypothesis class $\mathcal{F}$, we need # of samples that is:

- Finite Hypothesis class: $O(\log(|\mathcal{F}|)/\epsilon^2)$.
- Infinite hypothesis classes: $O(\text{VCdim}(\mathcal{F})/\epsilon^2)$.
- Linear Regression in d dimensions: $O(d/\epsilon^2)$

Generalization is possible in the IID supervised learning setting!!

To get ϵ -close to best in hypothesis class $\mathcal{F}$, we need # of samples that is:

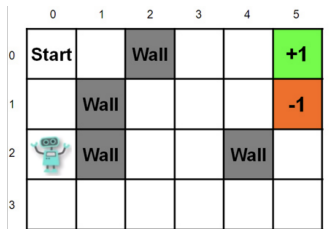
- Finite Hypothesis class: $O(\log(|\mathcal{F}|)/\epsilon^2)$.
- Infinite hypothesis classes: $O(\text{VCdim}(\mathcal{F})/\epsilon^2)$.
- Linear Regression in d dimensions: $O(d/\epsilon^2)$

The key idea in SL: uniform convergence / data-reuse.

With a training set, we can simultaneously evaluate the loss of all hypotheses in our class!

Sample Efficient RL in the Tabular Case (no generalization here)

Can we find an ϵ -opt policy with $\text{poly}(\mathcal{S}, \mathcal{A}, H, 1/\epsilon)$ samples?



Sample Efficient RL in the Tabular Case (no generalization here)

Can we find an ϵ -opt policy with $\text{poly}(\mathcal{S}, \mathcal{A}, H, 1/\epsilon)$ samples?

	0	1	2	3	4	5
0	Start		Wall			+1
1		Wall				-1
2		Wall			Wall	
3						

Theorem (Kearns & Singh '98; ...)

In the episodic setting, $\text{poly}(\mathcal{S}, \mathcal{A}, H, 1/\epsilon)$ samples suffice to find an ϵ -opt policy.

Sample Efficient RL in the Tabular Case (no generalization here)

Can we find an ϵ -opt policy with $\text{poly}(\mathcal{S}, \mathcal{A}, H, 1/\epsilon)$ samples?

	0	1	2	3	4	5
0	Start		Wall			+1
1		Wall				-1
2		Wall			Wall	
3						

Theorem (Kearns & Singh '98; ...)

In the episodic setting, $\text{poly}(\mathcal{S}, \mathcal{A}, H, 1/\epsilon)$ samples suffice to find an ϵ -opt policy.

- Key Idea: optimism + dynamic programming
- Add bonus for states which are not explored enough.

Q1: Can we find an ϵ -opt policy with no $|\mathcal{S}|$ dependence?

Chess has $|\mathcal{S}| \approx 10^{123}$
Dota2 has $\mathcal{S} \subset \mathbb{R}^{16000}!!$



Q1: Can we find an ϵ -opt policy with no $|S|$ dependence?

Chess has $|S| \approx 10^{123}$
Dota2 has $S \subset \mathbb{R}^{16000}!!$



- How can we reuse data to estimate the value of all policies in a policy class $\mathcal{F}$?

Q1: Can we find an ϵ -opt policy with no $|S|$ dependence?

Chess has $|S| \approx 10^{123}$
Dota2 has $S \subset \mathbb{R}^{16000}!!$



- How can we reuse data to estimate the value of all policies in a policy class $\mathcal{F}$?
Idea: Trajectory tree algorithm acts randomly for length H episodes and then uses importance sampling to evaluate every $f \in \mathcal{F}$.

Theorem (Kearns, Mansour, & Ng '00)

To find an ϵ -best in class policy, the trajectory tree algo uses $O(|\mathcal{A}|^H \log(|\mathcal{F}|)/\epsilon^2)$.

Q1: Can we find an ϵ -opt policy with no $|S|$ dependence?

Chess has $|S| \approx 10^{123}$
Dota2 has $S \subset \mathbb{R}^{16000}!!$



- How can we reuse data to estimate the value of all policies in a policy class $\mathcal{F}$?
Idea: Trajectory tree algorithm acts randomly for length H episodes and then uses importance sampling to evaluate every $f \in \mathcal{F}$.

Theorem (Kearns, Mansour, & Ng '00)

To find an ϵ -best in class policy, the trajectory tree algo uses $O(|\mathcal{A}|^H \log(|\mathcal{F}|)/\epsilon^2)$.

- Can we avoid A^H dependence to find an ϵ -best-in-class policy?

Q1: Can we find an ϵ -opt policy with no $|S|$ dependence?

Chess has $|S| \approx 10^{123}$
Dota2 has $S \subset \mathbb{R}^{16000}!!$



- How can we reuse data to estimate the value of all policies in a policy class $\mathcal{F}$?
Idea: Trajectory tree algorithm acts randomly for length H episodes and then uses importance sampling to evaluate every $f \in \mathcal{F}$.

Theorem (Kearns, Mansour, & Ng '00)

To find an ϵ -best in class policy, the trajectory tree algo uses $O(|A|^H \log(|\mathcal{F}|)/\epsilon^2)$.

- Can we avoid A^H dependence to find an ϵ -best-in-class policy?
Without further assumptions, NO!!
Proof: Consider a binary tree with 2^H policies and a sparse reward at a leaf node.

Q2: Can we find an ϵ -opt policy with no $|S|$, $|A|$ dependence
and $\text{poly}(H, 1/\epsilon, \text{"complexity measure"})$?

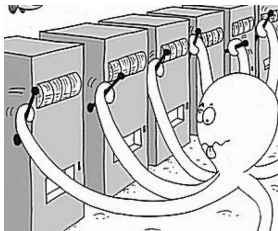
Q2: Can we find an ϵ -opt policy with no $|S|$, $|A|$ dependence and $\text{poly}(H, 1/\epsilon, \text{"complexity measure"})$?

- With various stronger assumptions, YES!
 - Linear Bellman Completion: [Munos et al. '05, Zanette et al. '19]
 - Linear MDPs: [Wang & Yang '18]; [Jin et al. '19] (the transition matrix is low rank)
 - Generalized Linear Bellman Completion: [Wang et al. '2019]
 - FLAMBE / Feature Selection: [Agarwal et al. '20]
 - Linear Mixture MDPs: [Modi et al. '20, Ayoub et al. '20]
 - Block MDPs [Du et al. '19]
 - Factored MDPs [Sun et al. '19]
 - Kernelized Nonlinear Regulator [Kakade et al. '20]
 - Linear Quadratic Regulators (LQR): standard control theory model
 - And more...

Part II: What are sufficient conditions for efficient RL?

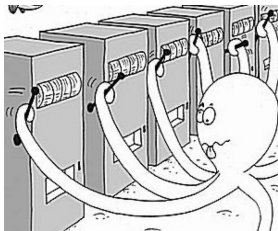
Is there a common theme to prior settings?

Warm Up: Linear bandits ($H = 1$ RL problem) [Abe and Long, 1999]



- [Assumption 1] One step RL ($H = 1$): single state: s_0 , large set of actions: $a \in \mathcal{A}$

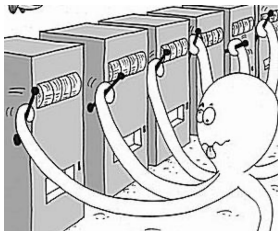
Warm Up: Linear bandits ($H = 1$ RL problem) [Abe and Long, 1999]



- [Assumption 1] One step RL ($H = 1$): single state: s_0 , large set of actions: $a \in \mathcal{A}$
- [Assumption 2] Linear reward: There exists unknown vector $w^* \in \mathbb{R}^d$ and known feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$

$$\mathbb{E}[r(s_0, a)] = \langle w^*, \phi(s_0, a) \rangle$$

Warm Up: Linear bandits ($H = 1$ RL problem) [Abe and Long, 1999]



- [Assumption 1] One step RL ($H = 1$): single state: s_0 , large set of actions: $a \in \mathcal{A}$
- [Assumption 2] Linear reward: There exists unknown vector $w^* \in \mathbb{R}^d$ and known feature map $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$

$$\mathbb{E}[r(s_0, a)] = \langle w^*, \phi(s_0, a) \rangle$$

Polynomial sample complexity is possible here [Auer et al. 2002; Dani et al. 2008]

Warm Up: Important structural property

- Linear “value-based” Hypothesis class $\mathcal{F}$:
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Warm Up: Important structural property

- Linear “value-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,

Warm Up: Important structural property

- Linear “value-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,

(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

Warm Up: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$
Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,
(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

An important structural property:

- **Bilinear Regret:** for all $w \in \mathcal{F}$, on policy difference between claimed reward $\mathbb{E}[Q_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\mathbb{E}_{\pi_w}[Q_w(s_0, a) - r]$$

Warm Up: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$
Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,
(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

An important structural property:

- **Bilinear Regret:** for all $w \in \mathcal{F}$, on policy difference between claimed reward $\mathbb{E}[Q_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_0, a) - r] \\ &= \mathbb{E}_{\pi_w} \left[\langle w, \phi(s_0, a) \rangle - \langle w^*, \phi(s_0, a) \rangle \right] \end{aligned}$$

Warm Up: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$
Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,
(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

An important structural property:

- **Bilinear Regret:** for all $w \in \mathcal{F}$, on policy difference between claimed reward $\mathbb{E}[Q_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_0, a) - r] \\ &= \mathbb{E}_{\pi_w} \left[\langle w, \phi(s_0, a) \rangle - \langle w^*, \phi(s_0, a) \rangle \right] \\ &= \langle w - w^*, \mathbb{E}_{\pi_w}[\phi(s_0, a)] \rangle \end{aligned}$$

Warm Up: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$
Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,
(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

An important structural property:

- **Bilinear Regret:** for all $w \in \mathcal{F}$, on policy difference between claimed reward $\mathbb{E}[Q_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned}\mathbb{E}_{\pi_w}[Q_w(s_0, a) - r] \\ &= \mathbb{E}_{\pi_w}[\langle w, \phi(s_0, a) \rangle - \langle w^*, \phi(s_0, a) \rangle] \\ &= \langle w - w^*, \mathbb{E}_{\pi_w}[\phi(s_0, a)] \rangle\end{aligned}$$

- **Data reuse:** There exists loss function $\ell(s, a, r, w') = Q_{w'}(s, a) - r$ such that the bilinear form for any hypothesis w' is estimable when playing π_w

Warm Up: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s_0, a) = \langle w, \phi(s_0, a) \rangle$,

(greedy) value $V_w(s_0)$ and (greedy) policy $\pi_w(s_0)$

An important structural property:

- **Bilinear Regret:** for all $w \in \mathcal{F}$, on policy difference between claimed reward $\mathbb{E}[Q_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned}\mathbb{E}_{\pi_w}[Q_w(s_0, a) - r] \\ &= \mathbb{E}_{\pi_w}[\langle w, \phi(s_0, a) \rangle - \langle w^*, \phi(s_0, a) \rangle] \\ &= \langle w - w^*, \mathbb{E}_{\pi_w}[\phi(s_0, a)] \rangle\end{aligned}$$

- **Data reuse:** There exists loss function $\ell(s, a, r, w') = Q_{w'}(s, a) - r$ such that the bilinear form for **any hypothesis w'** is estimable when playing π_w

$$\mathbb{E}_{\pi_w}[\ell(s_0, a, r, w')] = \langle w' - w^*, \mathbb{E}_{\pi_w}[\phi(s_0, a)] \rangle$$

Essentially, we can use data collected under π_w to estimate the bilinear form for w'

- Hypothesis class: $\{f \in \mathcal{F}\}$
with associated state action value $Q_f(s, a)$, (greedy) value $V_f(s)$ and (greedy) policy π_f
 - can be model-based or value-based class.

- Hypothesis class: $\{f \in \mathcal{F}\}$
with associated state action value $Q_f(s, a)$, (greedy) value $V_f(s)$ and (greedy) policy π_f
 - can be model-based or value-based class.

Definition

A $(\mathcal{F}, \ell)$ forms an (implicit) Bilinear class if there exists $w_h : \mathcal{F} \rightarrow \mathbb{R}^d$ and $\Phi_h : \mathcal{F} \rightarrow \mathbb{R}^d$ for all timesteps $h \in [H]$:

- Hypothesis class: $\{f \in \mathcal{F}\}$
with associated state action value $Q_f(s, a)$, (greedy) value $V_f(s)$ and (greedy) policy π_f
 - can be model-based or value-based class.

Definition

A $(\mathcal{F}, \ell)$ forms an (implicit) Bilinear class if there exists $w_h : \mathcal{F} \rightarrow \mathbb{R}^d$ and $\Phi_h : \mathcal{F} \rightarrow \mathbb{R}^d$ for all timesteps $h \in [H]$:

- **Bilinear regret:** on-policy difference between claimed reward and true reward satisfies a bilinear form:

$$|E_{\pi_f}[Q_f(s_h, a_h) - r(s_h, a_h) - V_f(s_{h+1})]| \leq |\langle w_h(f) - w_h(f^*), \Phi_h(f) \rangle|$$

- Hypothesis class: $\{f \in \mathcal{F}\}$
with associated state action value $Q_f(s, a)$, (greedy) value $V_f(s)$ and (greedy) policy π_f
 - can be model-based or value-based class.

Definition

A $(\mathcal{F}, \ell)$ forms an (implicit) Bilinear class if there exists $w_h : \mathcal{F} \rightarrow \mathbb{R}^d$ and $\Phi_h : \mathcal{F} \rightarrow \mathbb{R}^d$ for all timesteps $h \in [H]$:

- **Bilinear regret:** on-policy difference between claimed reward and true reward satisfies a bilinear form:

$$\left| E_{\pi_f} [Q_f(s_h, a_h) - r(s_h, a_h) - V_f(s_{h+1})] \right| \leq |\langle w_h(f) - w_h(f^*), \Phi_h(f) \rangle|$$

- **Data reuse:** There exists loss function $\ell_f(s_h, a_h, r_h, s_{h+1}, g)$ such that the bilinear form for any hypothesis g is estimable when playing π_f

$$\left| E_{\pi_f} [\ell_f(r_h, s_h, a_h, s_{h+1}, g)] \right| = |\langle w_h(g) - w_h(f^*), \Phi_h(f) \rangle|$$

Theorem 1: Structural Commonalities and Bilinear Classes

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

The following models are bilinear classes for some bounded discrepancy function $\ell(\cdot)$

- *Linear Bellman Completion*: [Munos et al. '05, Zanette et al. '19]
 - *Linear MDPs*: [Wang & Yang '18]; [Jin et al. '19] (the transition matrix is low rank)
 - *Generalized Linear Bellman Completion*: [Wang et al. '2019]
 - *FLAMBE / Feature Selection*: [Agarwal et al. '20]
 - *Linear Mixture MDPs*: [Modi et al. '20, Ayoub et al. '20]
 - *Block MDPs* [Du et al. '19]
 - *Factored MDPs* [Sun et al. '19]
 - *Kernelized Nonlinear Regulator* [Kakade et al. '20]
 - *Linear Quadratic Regulators (LQR)*: standard control theory model
 - And more...
-
- (almost) all “named” models (with provable generalization) are bilinear classes

Theorem 1: Structural Commonalities and Bilinear Classes

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

The following models are bilinear classes for some bounded discrepancy function $\ell(\cdot)$

- *Linear Bellman Completion*: [Munos et al. '05, Zanette et al. '19]
 - *Linear MDPs*: [Wang & Yang '18]; [Jin et al. '19] (the transition matrix is low rank)
 - *Generalized Linear Bellman Completion*: [Wang et al. '2019]
 - *FLAMBE / Feature Selection*: [Agarwal et al. '20]
 - *Linear Mixture MDPs*: [Modi et al. '20, Ayoub et al. '20]
 - *Block MDPs* [Du et al. '19]
 - *Factored MDPs* [Sun et al. '19]
 - *Kernelized Nonlinear Regulator* [Kakade et al. '20]
 - *Linear Quadratic Regulators (LQR)*: standard control theory model
 - And more...
- (almost) all “named” models (with provable generalization) are bilinear classes
two exceptions: a) deterministic linear Q^* [Wen & Van Roy, '13; Du, Lee, **M.**, Wang, '20]
b) Q^* state-action aggregation [Dong et al. '20]

Theorem 1: Structural Commonalities and Bilinear Classes

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

The following models are bilinear classes for some bounded discrepancy function $\ell(\cdot)$

- *Linear Bellman Completion*: [Munos et al. '05, Zanette et al. '19]
 - *Linear MDPs*: [Wang & Yang '18]; [Jin et al. '19] (the transition matrix is low rank)
 - *Generalized Linear Bellman Completion*: [Wang et al. '2019]
 - *FLAMBE / Feature Selection*: [Agarwal et al. '20]
 - *Linear Mixture MDPs*: [Modi et al. '20, Ayoub et al. '20]
 - *Block MDPs* [Du et al. '19]
 - *Factored MDPs* [Sun et al. '19]
 - *Kernelized Nonlinear Regulator* [Kakade et al. '20]
 - *Linear Quadratic Regulators (LQR)*: standard control theory model
 - And more...
-
- (almost) all “named” models (with provable generalization) are bilinear classes
 - two exceptions: a) deterministic linear Q^* [Wen & Van Roy, '13; Du, Lee, **M.**, Wang, '20]
b) Q^* state-action aggregation [Dong et al. '20]
 - Bilinear classes generalize the: Bellman rank [Jiang et al. '17]; Witness rank [Wen et al. '19]

Theorem 1: Structural Commonalities and Bilinear Classes

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

The following models are bilinear classes for some bounded discrepancy function $\ell(\cdot)$

- *Linear Bellman Completion*: [Munos et al. '05, Zanette et al. '19]
 - *Linear MDPs*: [Wang & Yang '18]; [Jin et al. '19] (the transition matrix is low rank)
 - *Generalized Linear Bellman Completion*: [Wang et al. '2019]
 - *FLAMBE / Feature Selection*: [Agarwal et al. '20]
 - *Linear Mixture MDPs*: [Modi et al. '20, Ayoub et al. '20]
 - *Block MDPs* [Du et al. '19]
 - *Factored MDPs* [Sun et al. '19]
 - *Kernelized Nonlinear Regulator* [Kakade et al. '20]
 - *Linear Quadratic Regulators (LQR)*: standard control theory model
 - And more. . .
-
- (almost) all “named” models (with provable generalization) are bilinear classes
 - two exceptions: a) deterministic linear Q^* [Wen & Van Roy, '13; Du, Lee, **M.**, Wang, '20]
b) Q^* state-action aggregation [Dong et al. '20]
 - Bilinear classes generalize the: Bellman rank [Jiang et al. '17]; Witness rank [Wen et al. '19]
 - The framework easily leads to new models (see paper).

Algorithm 1: BiLin-UCB

- 1 **Input** number of iterations T , estimator function ℓ , batch size m , confidence radius R
- 2 **Initialize** discrepancy function $\sigma : \mathcal{F} \rightarrow \mathbb{R}$ as $\sigma^2(\cdot) = 0$
- 3 **for** iteration $t = 0, 1, \dots, T - 1$ **do**

Algorithm 1: BiLin-UCB

- 1 **Input** number of iterations T , estimator function ℓ , batch size m , confidence radius R
- 2 **Initialize** discrepancy function $\sigma : \mathcal{F} \rightarrow \mathbb{R}$ as $\sigma^2(\cdot) = 0$
- 3 **for** iteration $t = 0, 1, \dots, T - 1$ **do**
- 4 **Find the optimistic $f_t \in \mathcal{F}$:**

$$f_t := \arg \max_f V_f(s_0) \quad \text{subject to } \sigma^2(f) \leq R$$

Algorithm 1: BiLin-UCB

- 1 **Input** number of iterations T , estimator function ℓ , batch size m , confidence radius R
- 2 **Initialize** discrepancy function $\sigma : \mathcal{F} \rightarrow \mathbb{R}$ as $\sigma^2(\cdot) = 0$
- 3 **for** iteration $t = 0, 1, \dots, T - 1$ **do**
- 4 **Find the optimistic $f_t \in \mathcal{F}$:**

$$f_t := \arg \max_f V_f(s_0) \quad \text{subject to } \sigma^2(f) \leq R$$

- 5 Sample m trajectories using π_{f_t} and create a batch dataset of size mH :

$$S = \{(r_h, s_h, a_h, s_{h+1}) \in \text{trajectories}\}$$

Algorithm 1: BiLin-UCB

- 1 **Input** number of iterations T , estimator function ℓ , batch size m , confidence radius R
 - 2 **Initialize** discrepancy function $\sigma : \mathcal{F} \rightarrow \mathbb{R}$ as $\sigma^2(\cdot) = 0$
 - 3 **for** iteration $t = 0, 1, \dots, T - 1$ **do**
 - 4 **Find the optimistic $f_t \in \mathcal{F}$:**
$$f_t := \arg \max_f V_f(s_0) \quad \text{subject to } \sigma^2(f) \leq R$$
 - 5 Sample m trajectories using π_{f_t} and create a batch dataset of size mH :
$$S = \{(r_h, s_h, a_h, s_{h+1}) \in \text{trajectories}\}$$
 - 6 Update the **discrepancy** function $\sigma^2(\cdot)$
$$\sigma^2(\cdot) \leftarrow \sigma^2(\cdot) + \left(\frac{1}{|S|} \sum_{o \in S} \ell(o, \cdot) \right)^2$$
 - 7 **return:** the best policy π_f found
-

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

Assume $(\mathcal{F}, \ell)$ is a bilinear class with $\Phi_h(f) \in \mathbb{R}^d$, bounded ℓ and the class is realizable, i.e.

$Q^* \in \mathcal{F}$. Using $\frac{d^2}{\epsilon^2} \cdot \text{poly}(H) \cdot \log(|\mathcal{F}|) \cdot \log(1/\delta)$ trajectories, the BiLin-UCB algorithm returns an ϵ -opt policy (with prob. $1 - \delta$).

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

Assume $(\mathcal{F}, \ell)$ is a bilinear class with $\Phi_h(f) \in \mathbb{R}^d$, bounded ℓ and the class is realizable, i.e.

$Q^* \in \mathcal{F}$. Using $\frac{d^2}{\epsilon^2} \cdot \text{poly}(H) \cdot \log(|\mathcal{F}|) \cdot \log(1/\delta)$ trajectories, the BiLin-UCB algorithm returns an ϵ -opt policy (with prob. $1 - \delta$).

- The proof is “elementary” using the elliptical potential function. [Dani et al., '08]

Theorem (Du, Kakade, Lee, Lovett, **M.**, Sun, Wang '21)

Assume $(\mathcal{F}, \ell)$ is a bilinear class with $\Phi_h(f) \in \mathbb{R}^d$, bounded ℓ and the class is realizable, i.e.

$Q^* \in \mathcal{F}$. Using $\frac{d^2}{\epsilon^2} \cdot \text{poly}(H) \cdot \log(|\mathcal{F}|) \cdot \log(1/\delta)$ trajectories, the BiLin-UCB algorithm returns an ϵ -opt policy (with prob. $1 - \delta$).

- The proof is “elementary” using the elliptical potential function. [Dani et al., '08]
- Extends to infinite dimensional problems using max info gain γ_T [Auer et al., '02; Srinivas et al., '10; Abbasi-Yadkori et al., '11]

- The proof follows from this lemma about **existence of high quality policy**.

Lemma (Existence of high quality policy)

Suppose we run the algorithm for $T \approx d$ iterations. Then, there exists $t \in [T]$ such that the following is true for hypothesis f_t :

$$V^* - V^{\pi_{f_t}}(s_0) \leq 2H\sqrt{d} \cdot \underbrace{H\sqrt{\frac{\log(|\mathcal{F}|)}{m}}}_{\text{SL generalization error of } \ell}$$

- **Bilinear regret assumption** and **Optimism** give an upper bound for sub-optimality.

Lemma (Bilinear Regret Lemma)

The following holds for all $t \in [T]$ w.h.p.:

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

- **Bilinear regret assumption** and **Optimism** give an upper bound for sub-optimality.

Lemma (Bilinear Regret Lemma)

The following holds for all $t \in [T]$ w.h.p.:

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

Proof:

$$V^*(s_0) - V^{\pi_{f_t}}(s_0)$$

- **Bilinear regret assumption** and **Optimism** give an upper bound for sub-optimality.

Lemma (Bilinear Regret Lemma)

The following holds for all $t \in [T]$ w.h.p.:

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

Proof:

$$\begin{aligned} V^*(s_0) - V^{\pi_{f_t}}(s_0) \\ \leq V_{f_t}(s_0) - V^{\pi_{f_t}}(s_0) \end{aligned} \quad (\text{optimism})$$

- **Bilinear regret assumption** and **Optimism** give an upper bound for sub-optimality.

Lemma (Bilinear Regret Lemma)

The following holds for all $t \in [T]$ w.h.p.:

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

Proof:

$$\begin{aligned} & V^*(s_0) - V^{\pi_{f_t}}(s_0) \\ & \leq V_{f_t}(s_0) - V^{\pi_{f_t}}(s_0) && \text{(optimism)} \\ & = \sum_{h=0}^{H-1} \mathbb{E}_{a_0:h \sim \pi_{f_t}} [Q_{f_t}(s_h, a_h) - r(s_h, a_h) - Q_{f_t}(s_{h+1}, a_{h+1})] && \text{(telescoping sum)} \end{aligned}$$

- **Bilinear regret assumption** and **Optimism** give an upper bound for sub-optimality.

Lemma (Bilinear Regret Lemma)

The following holds for all $t \in [T]$ w.h.p.:

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle|.$$

Proof:

$$\begin{aligned} & V^*(s_0) - V^{\pi_{f_t}}(s_0) \\ & \leq V_{f_t}(s_0) - V^{\pi_{f_t}}(s_0) && \text{(optimism)} \\ & = \sum_{h=0}^{H-1} \mathbb{E}_{a_0:h \sim \pi_{f_t}} [Q_{f_t}(s_h, a_h) - r(s_h, a_h) - Q_{f_t}(s_{h+1}, a_{h+1})] && \text{(telescoping sum)} \\ & = \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| && \text{(bilinear regret assumption)} \end{aligned}$$

- **Bilinear regret assumption** and **Optimism** give an upper bound on sub-optimality for all iterations t .

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

- **Bilinear regret assumption** and **Optimism** give an upper bound on sub-optimality for all iterations t .

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| .$$

- Our goal then is to show existence of iteration $t \in [T]$ such that

$$\sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| \quad \text{is small}$$

- **Bilinear regret assumption** and **Optimism** give an upper bound on sub-optimality for all iterations t .

$$V^* - V^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle|.$$

- Our goal then is to show existence of iteration $t \in [T]$ such that

$$\sum_{h=0}^{H-1} |\langle w_h(f_t) - w_h(f^*), \Phi_h(f_t) \rangle| \quad \text{is small}$$

- To that end, we will show existence of iteration $t \in [T]$ such that for $\Sigma_{0;h} = \lambda I$ and $\Sigma_{t;h} = \Sigma_{0;h} + \sum_{i=0}^{t-1} \Phi_h(f_i) \Phi_h(f_i)^\top$, the following is true

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \quad \|\Phi_h(f_t)\|_{\Sigma_{t;h}^{-1}} \quad \text{is small for all } h \in [H]$$

- To that end, we will show existence of iteration $t \in [T]$ such that for $\Sigma_{0;h} = \lambda I$ and $\Sigma_{t;h} = \Sigma_{0;h} + \sum_{i=0}^{t-1} \Phi_h(f_i) \Phi_h(f_i)^\top$, the following is true

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \quad \|\Phi_h(f_t)\|_{\Sigma_{t;h}^{-1}} \quad \text{is small for all } h \in [H]$$

- To that end, we will show existence of iteration $t \in [T]$ such that for $\Sigma_{0;h} = \lambda I$ and $\Sigma_{t;h} = \Sigma_{0;h} + \sum_{i=0}^{t-1} \Phi_h(f_i) \Phi_h(f_i)^\top$, the following is true

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \quad \|\Phi_h(f_t)\|_{\Sigma_{t;h}^{-1}} \quad \text{is small for all } h \in [H]$$

- From our **optimization constraint**, we get that for all time t (we can set R small because of **uniform convergence** and **Data reuse assumption**)

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \leq R = 2\sqrt{d} \cdot \underbrace{H \sqrt{\frac{\log(|\mathcal{F}|)}{m}}}_{\text{SL generalization error}} \quad \text{for all } h \in [H]$$

- To that end, we will show existence of iteration $t \in [T]$ such that for $\Sigma_{0;h} = \lambda I$ and $\Sigma_{t;h} = \Sigma_{0;h} + \sum_{i=0}^{t-1} \Phi_h(f_i) \Phi_h(f_i)^\top$, the following is true

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \quad \|\Phi_h(f_t)\|_{\Sigma_{t;h}^{-1}} \quad \text{is small for all } h \in [H]$$

- From our **optimization constraint**, we get that for all time t (we can set R small because of **uniform convergence** and **Data reuse assumption**)

$$\|w_h(f_t) - w_h(f^*)\|_{\Sigma_{t;h}} \leq R = 2\sqrt{d} \cdot \underbrace{H \sqrt{\frac{\log(|\mathcal{F}|)}{m}}}_{\text{SL generalization error}} \quad \text{for all } h \in [H]$$

- From **Elliptical Potential Lemma**, there exists $t \in [T]$ (for $T \approx d$) such that

$$\|\Phi_h(f_t)\|_{\Sigma_{t;h}^{-1}}^2 = O(1) \quad \text{for all } h \in [H]$$

Note that for infinite dimensional spaces, we can use max info gain instead.

Lemma (Elliptical Potential Lemma; Dani et al., '08)

Consider any sequence of vectors $\{x_0, \dots, x_{T-1}\}$ where $x_i \in \mathcal{V}$ for some Hilbert space $\mathcal{V}$. Let $\lambda \in \mathbb{R}^+$. Denote $\Sigma_0 = \lambda I$ and $\Sigma_t = \Sigma_0 + \sum_{i=0}^{t-1} x_i x_i^\top$. We have that:

$$\min_{i \in [T]} \ln \left(1 + \|x_i\|_{\Sigma_i^{-1}}^2 \right) \leq \frac{1}{T} \ln \frac{\det(\Sigma_T)}{\det(\lambda I)}.$$

Lemma (Elliptical Potential Lemma; Dani et al., '08)

Consider any sequence of vectors $\{x_0, \dots, x_{T-1}\}$ where $x_i \in \mathcal{V}$ for some Hilbert space $\mathcal{V}$. Let $\lambda \in \mathbb{R}^+$. Denote $\Sigma_0 = \lambda I$ and $\Sigma_t = \Sigma_0 + \sum_{i=0}^{t-1} x_i x_i^\top$. We have that:

$$\min_{i \in [T]} \ln \left(1 + \|x_i\|_{\Sigma_i^{-1}}^2 \right) \leq \frac{1}{T} \ln \frac{\det(\Sigma_T)}{\det(\lambda I)}.$$

- **Proof:** By definition of Σ_t and **matrix determinant lemma**, we have:

$$\ln \det(\Sigma_{t+1}) = \ln \det(\Sigma_t) + \ln \left(1 + \|x_t\|_{\Sigma_t^{-1}}^2 \right).$$

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

Special case II: Linear Bellman complete classes [Munos, 2005]

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

- **[Assumption 2] Completeness:** Let $\mathcal{F}$ be the linear “value-based” hypothesis class. For every $w \in \mathcal{F}$, there exists $T(w) \in \mathcal{F}$ such that

$$\langle T(w), \phi(s, a) \rangle = r(s, a) + \mathbb{E}_{s' \sim P(s, a)} [\max_{a'} Q_w(s', a')]$$

Special case II: Linear Bellman complete classes [Munos, 2005]

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

- **[Assumption 2] Completeness:** Let $\mathcal{F}$ be the linear “value-based” hypothesis class. For every $w \in \mathcal{F}$, there exists $T(w) \in \mathcal{F}$ such that

$$\langle T(w), \phi(s, a) \rangle = r(s, a) + \mathbb{E}_{s' \sim P(s, a)} [\max_{a'} Q_w(s', a')]$$

Polynomial sample complexity is possible here [Zanette et al. 2020])

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\mathbb{E}_{\pi_w}[Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})]$$

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w}[\langle w, \phi(s, a) \rangle - \langle T(w), \phi(s, a) \rangle] \end{aligned}$$

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w}[\langle w, \phi(s, a) \rangle - \langle T(w), \phi(s, a) \rangle] \\ &= \langle w - T(w), \mathbb{E}_{\pi_w}[\phi(s, a)] \rangle \end{aligned}$$

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w}[\langle w, \phi(s, a) \rangle - \langle T(w), \phi(s, a) \rangle] \\ &= \langle w - T(w), \mathbb{E}_{\pi_w}[\phi(s, a)] \rangle \\ &= \langle w - T(w) - (w^* - T(w^*)), \mathbb{E}_{\pi_w}[\phi(s, a)] \rangle \end{aligned}$$

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} [\langle w, \phi(s, a) \rangle - \langle T(w), \phi(s, a) \rangle] \\ &= \langle w - T(w), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \\ &= \langle w - T(w) - (w^* - T(w^*)), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell(\cdot, w')$ such that the bilinear form for **any hypothesis** w' is estimable when playing π_w

$$\begin{aligned} & \mathbb{E}_{\pi_w} [\ell(s_h, a_h, r_h, s_{h+1}, w')] \\ &= \langle w' - T(w') - (w^* - T(w^*)), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \end{aligned}$$

Special case II: Important structural property

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} [\langle w, \phi(s, a) \rangle - \langle T(w), \phi(s, a) \rangle] \\ &= \langle w - T(w), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \\ &= \langle w - T(w) - (w^* - T(w^*)), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell(\cdot, w')$ such that the bilinear form for **any hypothesis** w' is estimable when playing π_w

$$\begin{aligned} & \mathbb{E}_{\pi_w} [\ell(s_h, a_h, r_h, s_{h+1}, w')] \\ &= \langle w' - T(w') - (w^* - T(w^*)), \mathbb{E}_{\pi_w} [\phi(s, a)] \rangle \end{aligned}$$

Here the loss function is

$$\ell(s_h, a_h, r_h, s_{h+1}, w') = Q_{w'}(s_h, a_h) - r_h - V_{w'}(s_{h+1})$$

Linear Function Approximation

Basic idea: approximate the $Q(s, a)$ values with **linear basis functions** $\phi_1(s, a), \dots, \phi_d(s, a)$ (where $d \ll \#states, \#actions$).

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

Linear Function Approximation

Basic idea: approximate the $Q(s, a)$ values with **linear basis functions** $\phi_1(s, a), \dots, \phi_d(s, a)$ (where $d \ll \#states, \#actions$).

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

- **C. Shannon.** Programming a digital computer for playing chess. Philosophical Magazine, '50.

Linear Function Approximation

Basic idea: approximate the $Q(s, a)$ values with **linear basis functions** $\phi_1(s, a), \dots, \phi_d(s, a)$ (where $d \ll \#states, \#actions$).

- **[Assumption 1] Linear Q^* :** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle$$

- **C. Shannon.** Programming a digital computer for playing chess. Philosophical Magazine, '50.
- Lots of work on this approach, e.g. **TD-Gammon** [Tesauro, '95], **Atari** [Mnih+ '13].

Theorem (Weisz, Amortila, Szepesvári '21)

*There exists a **deterministic** MDP and ϕ satisfying **Assumption 1** s.t. any online RL algorithm requires $\Omega(\min(2^d, 2^H))$ samples to output optimal policy upto constant additive error.*

Theorem (Weisz, Amortila, Szepesvári '21)

There exists a **deterministic** MDP and ϕ satisfying **Assumption 1** s.t. any online RL algorithm requires $\Omega(\min(2^d, 2^H))$ samples to output optimal policy upto constant additive error.

- **[Assumption 2] Large Suboptimality Gap:** There is a $\Delta_{\min}$ such that for all $a \neq \pi^*(s)$

$$\inf_{s, a \neq \pi^*(s)} V_h^*(s) - Q_h^*(s, a) = \Delta_{\min} > 0$$

Theorem (Weisz, Amortila, Szepesvári '21)

There exists a **deterministic** MDP and ϕ satisfying **Assumption 1** s.t. any online RL algorithm requires $\Omega(\min(2^d, 2^H))$ samples to output optimal policy upto constant additive error.

- **[Assumption 2] Large Suboptimality Gap**: There is a $\Delta_{\min}$ such that for all $a \neq \pi^*(s)$

$$\inf_{s, a \neq \pi^*(s)} V_h^*(s) - Q_h^*(s, a) = \Delta_{\min} > 0$$

- **Efficient algorithms** exists for **deterministic** MDPs, stochastic rewards and **Assumption 1, 2** [Wen & Van Roy, '13; Du, Lee, **M.**, Wang, '20]

Theorem (Weisz, Amortila, Szepesvári '21)

There exists a **deterministic** MDP and ϕ satisfying **Assumption 1** s.t. any online RL algorithm requires $\Omega(\min(2^d, 2^H))$ samples to output optimal policy upto constant additive error.

- **[Assumption 2] Large Suboptimality Gap**: There is a $\Delta_{\min}$ such that for all $a \neq \pi^*(s)$

$$\inf_{s, a \neq \pi^*(s)} V_h^*(s) - Q_h^*(s, a) = \Delta_{\min} > 0$$

- **Efficient algorithms** exists for **deterministic** MDPs, stochastic rewards and **Assumption 1, 2** [Wen & Van Roy, '13; Du, Lee, **M.**, Wang, '20]

Theorem (Wang, Wang, Kakade '21)

There exists a **stochastic** MDP and ϕ satisfying **Assumption 1, 2** s.t. any online RL algorithm requires $\Omega(\min(2^d, 2^H))$ samples to output optimal policy upto constant additive error.

Special case III: Linear Q^* and Linear V^*

- [Assumption 1] Linear Q^* and V^* : There exists unknown $w^* \in \mathbb{R}^d$ and known features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle \quad \text{and} \quad V^*(s) = \langle w^*, \psi(s) \rangle$$

Special case III: Linear Q^* and Linear V^*

- [Assumption 1] Linear Q^* and V^* : There exists unknown $w^* \in \mathbb{R}^d$ and known features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ such that

$$Q^*(s, a) = \langle w^*, \phi(s, a) \rangle \quad \text{and} \quad V^*(s) = \langle w^*, \psi(s) \rangle$$

Can we get polynomial sample complexity
by also assuming linear V^* ?

Special case III: Important structural property

- Linear “value-based” Hypothesis class $\mathcal{F}$:
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Special case III: Important structural property

- Linear “value-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,

Special case III: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,
 $\pi_w(s)$ as the optimal functions for value function $Q_w(s, a)$

Special case III: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,

$\pi_w(s)$ as the optimal functions for value function $Q_w(s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})]$$

Special case III: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,

$\pi_w(s)$ as the optimal functions for value function $Q_w(s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \quad \mathbb{E}_{\pi_w} [\phi(s_h, a_h), -\psi(s_{h+1})] \right\rangle \end{aligned}$$

Special case III: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,

$\pi_w(s)$ as the optimal functions for value function $Q_w(s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \mathbb{E}_{\pi_w} [\phi(s_h, a_h), -\psi(s_{h+1})] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for **any hypothesis w'** is estimable when playing π_w

$$\begin{aligned} & \mathbb{E}_{\pi_w} [\ell_w(s_h, a_h, r_h, s_{h+1}, w')] \\ &= \left\langle w' - w^*, \mathbb{E}_{\pi_w} [\phi(s_h, a_h), -\psi(s_{h+1})] \right\rangle \end{aligned}$$

Special case III: Important structural property

- **Linear “value-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $Q_w(s, a) = \langle w, \phi(s, a) \rangle$, $V_w(s) = \langle w, \psi(s) \rangle$,

$\pi_w(s)$ as the optimal functions for value function $Q_w(s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w}[Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \mathbb{E}_{\pi_w}[\phi(s_h, a_h), -\psi(s_{h+1})] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for **any hypothesis w'** is estimable when playing π_w

$$\begin{aligned} & \mathbb{E}_{\pi_w}[\ell_w(s_h, a_h, r_h, s_{h+1}, w')] \\ &= \left\langle w' - w^*, \mathbb{E}_{\pi_w}[\phi(s_h, a_h), -\psi(s_{h+1})] \right\rangle \end{aligned}$$

Here the loss function is

$$\ell_w(s_h, a_h, r_h, s_{h+1}, w') = Q_{w'}(s_h, a_h) - V_{w'}(s_{h+1}) - r_h$$

- **[Assumption 1] Linear dynamics and rewards:** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$P(s' \mid s, a) = \langle w^*, \phi(s, a, s') \rangle \quad \text{and} \quad \mathbb{E}[r(s, a)] = \langle w^*, \psi(s, a) \rangle$$

- **[Assumption 1] Linear dynamics and rewards:** There exists **unknown** $w^* \in \mathbb{R}^d$ and **known** features $\phi : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ such that

$$P(s' \mid s, a) = \langle w^*, \phi(s, a, s') \rangle \quad \text{and} \quad \mathbb{E}[r(s, a)] = \langle w^*, \psi(s, a) \rangle$$

Polynomial sample complexity is possible here [Modi et al., 2020; Ayoub et al., 2020])

Special case IV: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:
set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Special case IV: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,

Special case IV: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,
 $Q_w(s, a)$, $V_w(s)$ and $\pi_w(s)$ as the optimal functions for model P_w

Special case IV: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,
 $Q_w(s, a)$, $V_w(s)$ and $\pi_w(s)$ as the optimal functions for model P_w

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})]$$

Special case IV: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,

$Q_w(s, a)$, $V_w(s)$ and $\pi_w(s)$ as the optimal functions for model P_w

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \quad \mathbb{E}_{\pi_w} \left[\psi(s_h, a_h) + \sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right] \right\rangle \end{aligned}$$

Special case IV: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,

$Q_w(s, a)$, $V_w(s)$ and $\pi_w(s)$ as the optimal functions for model P_w

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \mathbb{E}_{\pi_w} \left[\psi(s_h, a_h) + \sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for **any hypothesis w'** is estimable when playing π_w

$$\mathbb{E}_{\pi_w} [\ell(s_h, a_h, r_h, s_{h+1}, w')] = \left\langle w' - w^*, \mathbb{E}_{\pi_w} \left[\sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right] \right\rangle$$

Special case IV: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{w \in \mathbb{R}^d\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle w, \phi(s, a, s') \rangle$,

$Q_w(s, a)$, $V_w(s)$ and $\pi_w(s)$ as the optimal functions for model P_w

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - V_w(s_{h+1})] \\ &= \left\langle w - w^*, \mathbb{E}_{\pi_w} \left[\psi(s_h, a_h) + \sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for any hypothesis w' is estimable when playing π_w

$$\mathbb{E}_{\pi_w} [\ell(s_h, a_h, r_h, s_{h+1}, w')] = \left\langle w' - w^*, \mathbb{E}_{\pi_w} \left[\sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right] \right\rangle$$

Here the loss function is

$$\ell_w(s_h, a_h, r_h, s_{h+1}, w') = w_h'^\top \left(\psi(s_h, a_h) + \sum_{\bar{s} \in \mathcal{S}} \phi(s_h, a_h, \bar{s}) V_w(\bar{s}) \right) - V_w(s_{h+1}) - r_h$$

Special case V: Feature Selection for Low Rank MDPs

- [Assumption 1] Low rank MDP: There exists unknown features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ such that

$$P^*(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$$

Special case V: Feature Selection for Low Rank MDPs

- **[Assumption 1] Low rank MDP:** There exists **unknown** features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$, $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ such that

$$P^*(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$$

Polynomial sample complexity is possible here [Agarwal et al. 2020]

Special case V: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:
set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Special case V: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

Special case V: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

$Q_w(s, a)$, $V_w(s)$, $\pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Special case V: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

$Q_w(s, a)$, $V_w(s), \pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\mathbb{E}_{\pi_w} [Q_w(s_h, a_h)) - r(s_h, a_h) - \mathbb{E}V_w(s_{h+1})]$$

Special case V: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

$Q_w(s, a)$, $V_w(s), \pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - \mathbb{E}V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} \int_s (\mu^*(s))^\top \phi^*(s_{h-1}, a_{h-1}) [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds \end{aligned}$$

Special case V: Important structural property

- Linear “model-based” Hypothesis class $\mathcal{F}$:

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s') \rangle$,

$Q_w(s, a)$, $V_w(s), \pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Analogous structural property holds here:

- Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - \mathbb{E}V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} \int_s (\mu^\star(s))^\top \phi^\star(s_{h-1}, a_{h-1}) [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds \\ &= \left\langle \int_s (\mu^\star(s))^\top [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds, \mathbb{E}_{\pi_w} [\phi^\star(s_{h-1}, a_{h-1})] \right\rangle \end{aligned}$$

Special case V: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

$Q_w(s, a)$, $V_w(s), \pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - \mathbb{E}V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} \int_s (\mu^*(s))^\top \phi^*(s_{h-1}, a_{h-1}) [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds \\ &= \left\langle \int_s (\mu^*(s))^\top [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds, \mathbb{E}_{\pi_w} [\phi^*(s_{h-1}, a_{h-1})] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for **any hypothesis w'** is estimable

Special case V: Important structural property

- **Linear “model-based” Hypothesis class $\mathcal{F}$:**

set of all (bounded) linear vectors $\mathcal{F} = \{(\phi, \psi) \in \Phi \times \Psi\}$

Define for each hypothesis $w \in \mathcal{F}$, $P_w(s'|s, a) = \langle \phi(s, a), \psi(s) \rangle$,

$Q_w(s, a)$, $V_w(s), \pi_w(s)$ as the optimal functions for model $P_w(s'|s, a)$

Analogous structural property holds here:

- **Bilinear Regret:** on policy difference between claimed reward $\mathbb{E}[Q_w - V_w]$ and true reward $\mathbb{E}[r]$ satisfies a bilinear form

$$\begin{aligned} & \mathbb{E}_{\pi_w} [Q_w(s_h, a_h) - r(s_h, a_h) - \mathbb{E}V_w(s_{h+1})] \\ &= \mathbb{E}_{\pi_w} \int_s (\mu^\star(s))^\top \phi^\star(s_{h-1}, a_{h-1}) [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds \\ &= \left\langle \int_s (\mu^\star(s))^\top [V_w(s) - r(s, \pi_w(s)) - \mathbb{E}V_w(s')] ds, \mathbb{E}_{\pi_w} [\phi^\star(s_{h-1}, a_{h-1})] \right\rangle \end{aligned}$$

- **Data reuse:** There exists loss function $\ell_w(\cdot)$ such that the bilinear form for **any hypothesis w'** is estimable

$$\ell_w(s_h, a_h, r_h, s_{h+1}, w') = \frac{\mathbf{1}\{a_h = \pi_{w'}(s)\}}{1/A} (Q_{w'}(s_h, a_h) - r_h - V_{w'}(s_{h+1}))$$

Thanks!

- A generalization theory in RL is possible!
 - linear bandit theory $\rightarrow$ RL theory (bilinear classes) is rich.
 - covers known cases and new cases
 - leads to simple algorithm and proof
- Open Questions
 - Computational - Statistical Tradeoff.
 - Agnostic Realizable Equivalence



Simon Du



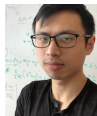
Sham Kakade



Jason Lee



Shachar Lovett



Wen Sun



Ruosong Wang